

Quick Check

Lesson 8: Final Explanation: Why Cassava Cuttings Grow Without Seeds but a Lone Pawpaw Needs a Partner Tree

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.1: Reproduction in Plants | Lesson 8

A lone avocado tree in a Nairobi compound is covered in flowers but sets no fruit. A second avocado tree is planted 10 m away. Which outcome is most likely once both trees flower?

- A** The second tree gives the first tree extra water, so its flowers will make fruit
- B** The first tree will start making fruit from its flowers by mitosis alone
- C** The two trees join their roots, so no pollen is needed to make fruit
- D** Pollen can be carried between the two trees, so cross-pollination and fertilisation can happen and fruit can form

A farmer in Kisumu plants a cassava stem cutting. It grows roots and shoots without any flower or pollen. Which statement explains this correctly?

- A** The cutting needs pollen from another cassava plant before it can grow roots
- B** The cutting is a type of seed, so it grows like a seed
- C** Cells in the stem divide by meiosis, so the new plant is different from the parent
- D** Cells in the stem divide by mitosis to form roots and shoots, so the new plant is a clone of the parent

Which of these is a feature that hinders self-pollination in flowering plants?

- A** Anthers and stigma of the same flower ripen at the same time and touch each other
- B** The flower produces a lot of nectar
- C** The flower has a large ovary with many ovules
- D** The stigma of a hibiscus flower matures at a different time from its anthers

During double fertilisation in a flowering plant such as maize, what happens to the two sperm nuclei from the pollen tube?

- A** One sperm nucleus forms the seed coat, and the other forms the fruit wall
- B** One sperm nucleus fuses with the egg cell to form the zygote, and the other fuses with the polar nuclei to form the endosperm
- C** Both sperm nuclei fuse with the polar nuclei to form the endosperm
- D** Both sperm nuclei fuse with the egg cell to form two zygotes

A learner cuts open a tomato fruit and finds small seeds inside. Which statement about how the fruit and seeds formed is correct?

- A** The petals became the fruit and the anthers became the seeds
- B** The ovule became the fruit and the ovary became the seeds
- C** The whole flower became the fruit and the seeds came from the stigma
- D** The ovary became the fruit and the ovules became the seeds

A student looks at a pawpaw flower and sees pollen grains on the stigma. Which conclusion is correct?

- A** Neither pollination nor fertilisation can happen without a second tree
- B** Pollination and fertilisation mean the same thing, so the fruit will form
- C** Pollination has happened, but fertilisation has not necessarily happened yet
- D** Fertilisation has definitely happened, because pollen is on the stigma

A sugarcane farmer in Mumias plants setts (stem pieces) instead of seeds. What is an advantage and a disadvantage of this method?

- A** Advantage: it needs double fertilisation. Disadvantage: it makes fewer plants
- B** Advantage: the crop grows fast and is uniform. Disadvantage: the plants are clones, so a disease can spread to all of them
- C** Advantage: it needs pollen from another farm. Disadvantage: it makes the plants different from each other
- D** Advantage: it creates a lot of genetic variation. Disadvantage: it needs pollinators

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).